Data Streaming: Crypto Market Data

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Data Streaming: Crypto Market Data

The Problem: High-velocity market streams cause API connection exhaustion, while traditional S3 data lakes suffer from "invisible data" requiring constant manual partition repairs.

The Solution: A fully serverless ingestion engine and Medallion Data Lake powered by Kinesis Firehose, AWS Glue, and Apache Iceberg for true ACID transactions.

The Benefit: Eliminates infrastructure overhead and unlocks instant SQL access to production-ready data with zero manual maintenance.

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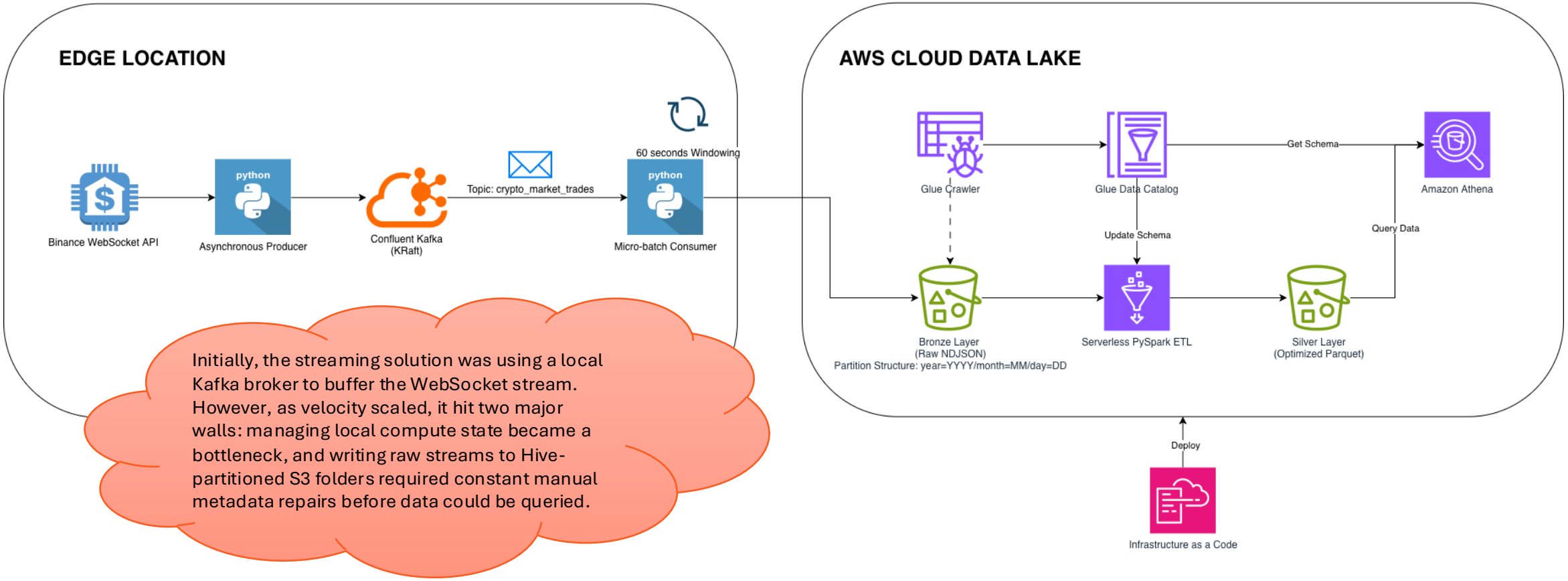
Local Kafka and Parquet Architecture (V1)

CRYPTO MARKET TRADE POC



Initially, the streaming solution was using a local Kafka broker to buffer the WebSocket stream. However, as velocity scaled, it hit two major walls: managing local compute state became a bottleneck, and writing raw streams to Hive-partitioned S3 folders required constant manual metadata repairs before data could be queried.

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The Architectural Pivot

Legacy State (Local Kafka & Parquet):

Compute: Relied on local hardware to run the message broker.

Storage: Required manual MSCK REPAIR TABLE commands to make newly streamed data visible to Amazon Athena.

Current State (Serverless & Iceberg):

Compute: Containerized Python producer deployed to AWS Fargate (0.25 vCPU) for 24/7 headless execution.

Ingestion: Kinesis Firehose buffers and micro-batches records to solve latency bottlenecks and eliminate broken pipes.

Storage: Apache Iceberg replaces folder-based tracking with file-level transaction logs, completely automating metadata management.

Orchestration: A native AWS Glue Scheduled Trigger automates the 15-minute Silver ETL micro-batches with zero cross-service IAM overhead.

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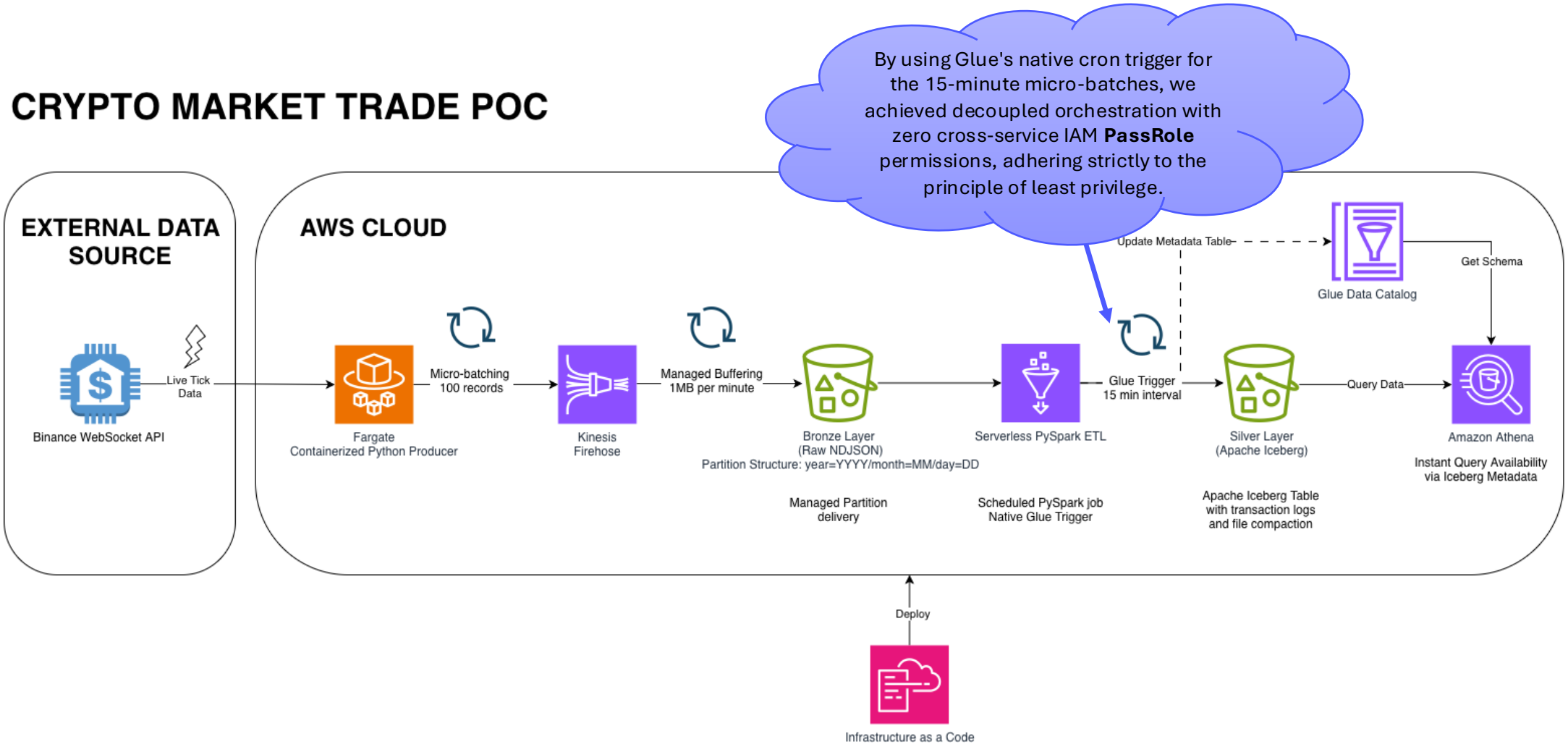
Serverless and Iceberg Architecture (V2)



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Outcome & Results

Cost Optimization: Fargate footprint minimized (0.25 vCPU) and ETL batched to 15-minute intervals, slashing compute overhead compared to real-time event-driven processing.

Zero Maintenance Operations: The combination of managed Firehose buffering and Apache Iceberg completely eliminated the need for infrastructure management and partition repairs.

Time-to-Insight: Data analysts can unlock instant SQL access to production-ready data with guaranteed schema enforcement immediately after a micro-batch completes.

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Links:

General API Reference:

- Binance Official Spot API Documentation: https://developers.binance.com/docs/binance-spot-api-docs

Specific WebSocket Reference (Best for this POC):

- Binance Official Market Data WebSocket Streams: https://developers.binance.com/docs/binance-spot-api-docs/web-socket-streams

GitHub Repo:

- GitHub: https://github.com/evan-gloria/crypto-market-data-pipeline

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